

# Minime CI Workflow Architecture (Vertical Pipeline)

Unified GitHub Actions Build System — `.github/workflows/build.yml` [\[Shared\]](#)

Legend: [\[Shared\]](#) = Dual-distro / shared across targets | [\[Specific\]](#) = Single-distro, board, or libc specific

## 1. Trigger / Entrypoint

- Workflow File: `.github/workflows/build.yml` [\[Shared\]](#)
- Triggers: Push to `main` (filtered to `minime/**`), Pull Request, or `workflow_dispatch`

↓ *(Triggers parallel asset build jobs)*

## 2. Parallel assets build - bootloader and UI

| 2A. Bootloader Build                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | 2B. UI Build (musl)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | 2C. UI Build (glibc)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>build-bootloader</b> (AMD64) <ul style="list-style-type: none"><li>• Workflow Job: <code>build-bootloader</code> <a href="#">[Shared]</a></li><li>• Runner: <code>ubuntu-latest</code></li><li>• Container: <code>minime-glibc:latest</code> <a href="#">[Shared]</a></li><li>• Script: <code>scripts/build-bootloader.sh</code> <a href="#">[Shared]</a></li><li>• Cache Key: <code>bootloader-*</code> (hash of <code>uboot/**</code>)</li><li>• Configs &amp; Patches:<ul style="list-style-type: none"><li>- <code>minime/uboot/config/uboot.config</code> <a href="#">[Shared]</a></li><li>- <code>minime/uboot/config/ddr3.defconfig</code> <a href="#">[Specific]</a> (H700)</li><li>- <code>minime/uboot/out/rk3566/rkbin/</code> <a href="#">[Specific]</a> (RK3566)</li></ul></li><li>• GH Artifact: <code>bootloader-out</code></li></ul> | <b>build-ui (musl)</b> (ARM64) <ul style="list-style-type: none"><li>• Workflow Job: <code>build-ui (musl)</code> <a href="#">[Shared]</a></li><li>• Runner: <code>ubuntu-24.04-arm</code></li><li>• Container: <code>minime-musl:latest</code> <a href="#">[Shared]</a></li><li>• Script: <code>scripts/build-ui.sh</code> <a href="#">[Shared]</a></li><li>• Cache Key: <code>ui-musl-*</code> (hash of <code>ui/**</code>)</li><li>• Submodules:<ul style="list-style-type: none"><li>- <code>minime/ui/minui/</code> <a href="#">[Shared]</a> (dual <code>musl/glibc</code>)</li><li>- <code>minime/ui/allium/</code> <a href="#">[Shared]</a> (dual <code>musl/glibc</code>)</li></ul></li><li>• GH Artifact: <code>ui-musl</code><ul style="list-style-type: none"><li>■ <code>minui-musl-aarch64.tar.xz</code></li><li>■ <code>allium-musl-aarch64.tar.xz</code></li></ul></li></ul> | <b>build-ui (glibc)</b> (AMD64) <ul style="list-style-type: none"><li>• Workflow Job: <code>build-ui (glibc)</code> <a href="#">[Shared]</a></li><li>• Runner: <code>ubuntu-latest</code></li><li>• Container: <code>minime-glibc:latest</code> <a href="#">[Shared]</a></li><li>• Script: <code>scripts/build-ui.sh</code> <a href="#">[Shared]</a></li><li>• Cache Key: <code>ui-glibc-*</code> (hash of <code>ui/**</code>)</li><li>• Submodules:<ul style="list-style-type: none"><li>- <code>minime/ui/minui/</code> <a href="#">[Shared]</a> (dual <code>musl/glibc</code>)</li><li>- <code>minime/ui/allium/</code> <a href="#">[Shared]</a> (dual <code>musl/glibc</code>)</li></ul></li><li>• GH Artifact: <code>ui-glibc</code><ul style="list-style-type: none"><li>■ <code>minui-glibc-aarch64.tar.xz</code></li><li>■ <code>allium-glibc-aarch64.tar.xz</code></li></ul></li></ul> |

↓ *(Uploads asset artifacts to GH & Passes to OS build matrix)*

## 3. Artifact Transfer & Caching Layer

- Actions Used: `actions/download-artifact@v4` [\[Shared\]](#), `actions/cache/restore@v4` [\[Shared\]](#)
- Artifacts Downloaded: `bootloader-out` + matching UI artifact (`ui-musl` for Alpine, `ui-glibc` for Buildroot)
- Asset Output Caches: Bootloader Cache (`bootloader-*`), UI Cache (`ui-musl-*`, `ui-glibc-*`)
- OS Compiler Caches: CCache (`~/alpine-ccache`, `~/buildroot-ccache`) + DL Cache (`~/alpine-dl`, `~/buildroot-dl`) [\[Specific\]](#)

↓ *(Fans out to 6 parallel matrix jobs)*

## 4. Parallel components build - kernel, rootfs, initramfs and dtb (build.sh)

| 4A. Alpine Component Builder (musl)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | 4B. Buildroot Component Builder (glibc)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Alpine (musl) Component Builder (3 Jobs)</b> <ul style="list-style-type: none"><li>• Command: <code>make -C minime/targets/alpine components BOARD=&lt;board_name&gt;</code> <a href="#">[Specific]</a></li><li>• Builder Script: <code>minime/targets/alpine/scripts/build.sh</code> <a href="#">[Specific]</a></li><li>• Inputs: <code>minime/boards/&lt;board_name&gt;/</code> <a href="#">[Specific]</a>, <code>minime/boards/common/</code> <a href="#">[Shared]</a></li><li>• Produced Component Artifacts:<ul style="list-style-type: none"><li>■ <code>system.erofs</code> (Immutable EROFS rootfs with Mesa/Panfrost)</li><li>■ <code>Image</code> (Linux kernel binary)</li><li>■ <code>initramfs &amp; *.dtb</code> (Device Tree Blobs)</li></ul></li></ul> | <b>Buildroot (glibc) Component Builder (3 Jobs)</b> <ul style="list-style-type: none"><li>• Command: <code>make -C minime/targets/buildroot components BOARD=&lt;board_name&gt;</code> <a href="#">[Specific]</a></li><li>• Builder Script: <code>minime/targets/buildroot/scripts/build.sh</code> <a href="#">[Specific]</a></li><li>• Inputs: <code>minime/boards/&lt;board_name&gt;/</code> <a href="#">[Specific]</a>, <code>minime/boards/common/</code> <a href="#">[Shared]</a></li><li>• Produced Component Artifacts:<ul style="list-style-type: none"><li>■ <code>system.erofs</code> (Immutable EROFS rootfs with glibc &amp; Mali drivers)</li><li>■ <code>Image</code> (Linux kernel binary)</li><li>■ <code>initramfs &amp; *.dtb</code> (Device Tree Blobs)</li></ul></li></ul> |

↓ *(Passes components (system.erofs, Image, DTBs) to Packaging)*

## 5. Image Assembly & Packaging (mkimage.sh / mkupdate.sh)

| 5A. Alpine Image Assembly (musl)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | 5B. Buildroot Image Assembly (glibc)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Alpine Image &amp; Update Packager (musl)</b> <ul style="list-style-type: none"><li>• Commands:<ul style="list-style-type: none"><li>- <code>make image BOARD=&lt;board_name&gt; UI=minui allium</code> <a href="#">[Specific]</a></li><li>- <code>make update BOARD=&lt;board_name&gt; UI=minui allium</code> <a href="#">[Specific]</a></li></ul></li><li>• Packaging Scripts:<ul style="list-style-type: none"><li>- <code>minime/build/genassets.sh</code> <a href="#">[Shared]</a></li><li>- <code>minime/build/mkimage.sh</code> <a href="#">[Shared]</a></li><li>- <code>minime/build/mkupdate.sh</code> <a href="#">[Shared]</a></li></ul></li><li>• Produced Release Archives:<ul style="list-style-type: none"><li>■ <code>*.img.xz</code> (Raw SD card flashable image)</li><li>■ <code>*.tar.xz</code> (OTA update package)</li></ul></li></ul> | <b>Buildroot Image &amp; Update Packager (glibc)</b> <ul style="list-style-type: none"><li>• Commands:<ul style="list-style-type: none"><li>- <code>make image BOARD=&lt;board_name&gt; UI=minui allium</code> <a href="#">[Specific]</a></li><li>- <code>make update BOARD=&lt;board_name&gt; UI=minui allium</code> <a href="#">[Specific]</a></li></ul></li><li>• Packaging Scripts:<ul style="list-style-type: none"><li>- <code>minime/build/genassets.sh</code> <a href="#">[Shared]</a></li><li>- <code>minime/build/mkimage.sh</code> <a href="#">[Shared]</a></li><li>- <code>minime/build/mkupdate.sh</code> <a href="#">[Shared]</a></li></ul></li><li>• Produced Release Archives:<ul style="list-style-type: none"><li>■ <code>*.img.xz</code> (Raw SD card flashable image)</li><li>■ <code>*.tar.xz</code> (OTA update package)</li></ul></li></ul> |

↓ *(Uploads output archives to GitHub Releases)*

## 6. Release & Delivery

- Workflow Step: Upload to Testing Release in `.github/workflows/build.yml` [\[Shared\]](#)
- Tool / CLI: `gh release upload testing <out>/*.img.xz <out>/*.tar.xz --clobber` [\[Shared\]](#)
- Destination: GitHub Releases — `testing` tag